

Elementary Problems Combinatorial Geometry

Version 1.0

10-12-2018

Basic Work

1. At most how many intersection points can be formed if n straight lines are drawn on the plane?
2. There are 2 points P and Q lying inside a pentagon. Prove that P and Q lie inside a quadrilateral formed by 4 of the vertices of the pentagon.
3. Let $\mathcal{P}$ be a convex polygon. Prove that a line ℓ not passing through any vertices of $\mathcal{P}$ intersects $\mathcal{P}$ in at most two distinct points.
4. Let n be a positive integer. There are n red points and n blue points on the circumference of a circle. Prove that we can use these $2n$ points to form n non-intersecting chords each of whose endpoints are of distinct colours.
5. (South Africa MO 2016 Problem 4) For which integers $n \geq 2$ is it possible to draw n straight lines in the plane in such a way that there are at least $n - 2$ points where exactly three of the lines meet?
6. (IMO shortlist 1966) What is the maximal number of regions a circle can be divided in by segments joining n points on the boundary of the circle?
7. (IMO shortlist 1972) We are given $3n$ points $A_1, A_2, \dots, A_{3n}$ in the plane, no three of them collinear. Prove that one can construct n disjoint triangles with vertices at the points A_i .
8. Let $\mathcal{P}$ be a convex polygon with $2n$ sides where $n \geq 2$ is a positive integer. Prove that there is a diagonal of $\mathcal{P}$ which is not parallel to any side of $\mathcal{P}$.
9. Let n be a positive integer. Prove that for any n^3 points on the plane with no three collinear, we can find n points such that there are no isosceles triangles among these n points.
10. All the points on a plane are coloured with three colours. Prove that there exist two points of the same colour whose distance is 1.
11. (India MO 2008 Problem 4) All the points with integer coordinates in the xy -plane are coloured using three colours, red, blue and green, each colour being used at least once. It is known that the point $(0,0)$ is red and the point $(0,1)$ is blue. Prove that there exist three points with integer coordinates of distinct colours which form the vertices of a right-angled triangle.
(Remark: The colours of $(0,0)$ and $(0,1)$ are unimportant. The right-angled triangle needs not have legs parallel to the coordinate axes.)
12. Let $n \geq 2$ be a positive integer. There is a set of n distinct points on the plane. At most how many axes of symmetry does this set have?

13. Let n be a positive integer. Prove that for any $2n$ points on the plane, there exists a circle (not necessarily formed by some of the $2n$ points) which contains exactly n of these points in its interior.
14. Let n be a positive integer. Let $\mathcal{A}$ be a set of $2n$ points on the plane such that no 3 points are collinear. Prove that for any pair of points $A, B \in \mathcal{A}$, there exists a line on the plane which separates the points in $\mathcal{A}$ into two halves such that one half contains n points including A , and another half contains n points including B .
15. (CMO 1991 Problem 3) There are 10 birds on the ground. Among any 5 birds, there are at least 4 of them which lie on the circumference of a circle. At least how many birds are there on the circumference of a circle with the greatest number of birds?

Geometric Facts

16. Prove that we can colour all the points on a circle red and blue such that there is no right-angled triangle inscribed in the circle whose 3 vertices are of the same colour.
17. (IMO shortlist 1966) Given a regular n -gon $A_1A_2 \cdots A_n$ (with $n \geq 3$) in a plane, how many triangles of the kind $A_iA_jA_k$ are obtuse?
18. A convex polygon with $n \geq 3$ sides is dissected into m triangles such that each side of each triangle is either a side of another triangle or a side of the polygon. Prove that $m + n$ is even.
19. There are finitely many points on the plane such that the distances between all pairs of points are distinct. From each point we draw a directed edge to its nearest point. Prove that no point is the terminal point of more than 5 directed edges.
20. (IMO shortlist 1967) The n points $P_1, P_2, \dots, P_n$ are placed inside or on the boundary of a disk of radius 1 in such a way that the minimum distance between any two of these points has its largest possible value D_n . Calculate D_n for $n = 2$ to 7 and justify your answer.
21. (France TST 2007 Problem 4) Do there exist 5 points in the space, such that for all $n \in \{1, 2, \dots, 10\}$ there exist two of them at distance between them n ?
22. (IMO 1965 Problem 6) In a plane a set of n points ($n \geq 3$) is given. Each pair of points is connected by a segment. Let d be the length of the longest of these segments. We define a diameter of the set to be any connecting segment of length d . Prove that the number of diameters of the given set is at most n .
23. (China TST 2007) Suppose there are $n \geq 3$ points in the plane. Prove that there exist three points A, B, C satisfying $1 \leq \frac{AB}{AC} \leq \frac{n+1}{n-1}$.

Convex Hull

24. (IMO shortlist 1966) Given 5 points in the plane, no three of them are collinear. Show that among these 5 points, we can always find 4 points forming a convex quadrilateral.
25. Prove that there is no function f mapping the points on a plane to the points on the plane such that for any convex quadrilateral $ABCD$, the images $f(A), f(B), f(C), f(D)$ are the vertices of a concave quadrilateral.
26. Given $n \geq 4$ points in the plane such that any 4 of them form a convex quadrilateral. Prove that the n points form a convex polygon with n sides.
27. (IMO shortlist 1966) Given $n > 3$ points in the plane such that no three of the points are collinear. Does there exist a circle passing through (at least) 3 of the given points and not containing any other of the n points in its interior?
28. Let n be a positive integer. There are $2n + 3$ points on the plane with no three collinear and no four concyclic. Prove that there exist 3 of the points whose circumcircle splits the other $2n$ points into two halves, with n points lying inside the circle and n points lying outside the circle.
29. There are n points on the plane with no three collinear. If there are only k different distances formed between pairs of the points, prove that $n \leq k^2 + 2$.
30. (Canada MO 2005 Problem 3) Let S be a set of $n \geq 3$ points in the interior of a circle.
 - (a) Show that there are three distinct points $a, b, c \in S$ and three distinct points A, B, C on the circle such that a is (strictly) closer to A than any other point in S , b is closer to B than any other point in S and c is closer to C than any other point in S .
 - (b) Show that for no value of n can four such points in S (and corresponding points on the circle) be guaranteed.
31. Let $n \geq 3$ be a positive integer. There are n points on the plane with no three collinear. We draw some segments using these n points as endpoints such that no two segments meet in the interior. Considering all positions of the n points, find the maximum number of segments that we can draw.
32. Let n be a positive integer. There are $4n$ points A_j on the plane and another point P such that no three of these $4n + 1$ points are collinear. Prove that there is an even number of triangles $A_i A_j A_k$ containing P in the interior.

Extremal Case

33. Let S be a set of points on the plane such that every point in S is the midpoint of two points in S . Prove that S is an infinite set.
34. Let n be a positive integer. There are n red points and n blue points on the plane with no three collinear. Prove that we can draw n segments such that the endpoints of each segment have distinct colours and any pair of segments do not meet each other (in the interior or at the endpoint).
35. Let S be a set of points on the plane with no three collinear. If any triangle with vertices from S has area at most 1, prove that S can be covered by a triangle with area 4.
36. (Balkan MO 2010 Problem 3) A strip of width w is the set of all points which lie on, or between, two parallel lines distance w apart. Let S be a set of n ($n \geq 3$) points on the plane such that any three different points of S can be covered by a strip of width 1. Prove that S can be covered by a strip of width 2.
37. (USATST 2005 Problem 5) Find all finite sets S of points in the plane with the following property: for any three distinct points A, B , and C in S , there is a fourth point D in S such that A, B, C , and D are the vertices of a parallelogram (in some order).
38. Let $A_1A_2 \cdots A_n$ be a convex polygon with $n \geq 3$ sides. Prove that there exists k such that the polygon lies inside or on the boundary of the circumcircle of $\triangle A_kA_{k+1}A_{k+2}$ (where $A_{n+1} = A_1$ and $A_{n+2} = A_2$).

Record

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